



# Enabling Onboard Navigation for UAVs in GPS Denied Environments

**MIP400A : B.Tech Project - MidTerm Evaluation**

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**Goal:** UAV self-localization when GPS/GNSS is unavailable, jammed, or spoofed, using only onboard sensors.

- ❑ **This semester:** working end-to-end pipeline, in simulation.
- ❑ **Next semester (through Apr 2027):** fast, power-efficient (FPGA) onboard deployment.

## Core approach: Scene-Aided Navigation

Replace GPS with passive measurements the aircraft can already sense: terrain, gravity, magnetic field, and visual scene, matched against a reference map.



## **Weak signal**

GNSS is easily jammed, spoofed, or simply unavailable.

## **Mechanical view**

The IMU already senses acceleration, rotation, and gravity; a camera could replace the satellite link entirely.

## **Must keep flying**

Survey, disaster-response, and defense UAVs need to operate even when GPS fails.

## **Precedent**

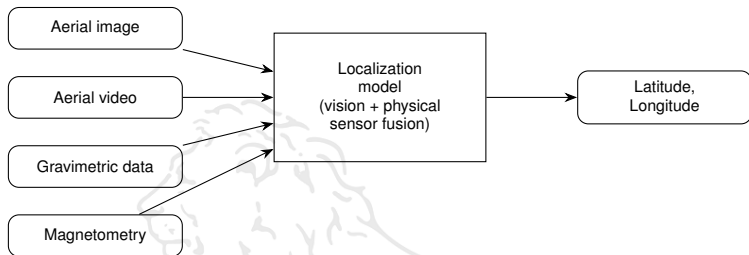
TERCOM and DSMAC proved passive, GPS-free terrain and scene matching decades ago.



Passive GPS-denied navigation methods, screened for feasibility on a small UAV:

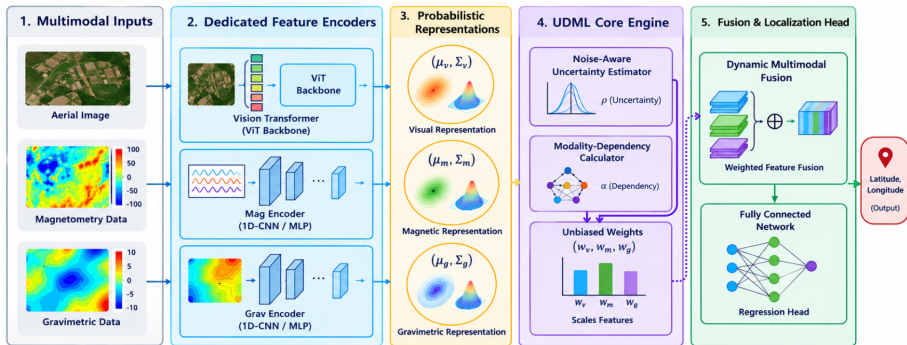
| Method                     | Physical quantity measured      | Passive? | Assessment                                                    |
|----------------------------|---------------------------------|----------|---------------------------------------------------------------|
| Terrain matching (TERCOM)  | Ground elevation                | Yes      | Proven, strong candidate <sup>[4,6]</sup>                     |
| Visual landmark matching   | Camera imagery                  | Yes      | Strong if paired with AI matching <sup>[1-3,5,7,12,13]</sup>  |
| Transformer-based matching | Camera imagery                  | Yes      | SOTA in this niche; motivates our redesign <sup>[14,15]</sup> |
| Gravity matching           | Gravity anomalies               | Yes      | Promising, needs sensitive sensor <sup>[8,9]</sup>            |
| Gravity-gradient nav.      | $\partial g / \partial x, y, z$ | Yes      | Very promising, higher precision <sup>[8,9]</sup>             |
| Magnetic anomaly nav.      | Magnetic field                  | Yes      | Useful in mountainous terrain <sup>[10,11]</sup>              |
| Quantum inertial sensing   | Acceleration / rotation         | Yes      | No map needed; pure mechanics                                 |
| Wi-Fi / cellular / RF      | Radio signal strength           | No       | Rejected: unavailable at altitude                             |

**Reviewed:** VGG16<sup>[1]</sup>, CLIP<sup>[2,3,7]</sup>, SIFT/ORB<sup>[5]</sup>, and DEM terrain-weighting<sup>[4,6]</sup>, achieving sub-3–7 m RMSE<sup>[2,4]</sup>.



- ❑ **Baseline validated:** replicated the reference implementation<sup>[1]</sup> in simulation; accuracy matches the paper's reported numbers.
- ❑ **Now:** redesigning the backbone as a **transformer** (next slide), trading compute and latency for higher accuracy, then adding multimodal sensor fusion.

# Proposed Model Architecture



- Transformer vision backbone (ViT) replaces VGG16<sup>[1]</sup>; magnetometry and gravimetry get their own 1D-CNN/MLP encoders.
- Each modality is encoded as a **probabilistic representation**  $(\mu, \Sigma)$ , so sensor noise is carried forward explicitly.
- The **UDML core** estimates per-modality uncertainty  $\rho$  and dependency  $\alpha$ , producing unbiased fusion weights  $(w_v, w_m, w_g)$ .
- Weighted features are fused and regressed to **latitude, longitude** by a fully connected head.

# Timeline of the Work



## Phase 1: Now to Nov 2026

Transformer backbone +  
multimodal fusion  
(IMU, magnetometer, gravimeter)  
*end-to-end pipeline, in simulation*

## Phase 2: Jan to Apr 2027

Optimize for speed  
& power on FPGA  
*physical UAV test*

Phase 2 extends our lab project from last semester, which showed FPGAs are highly power-efficient for this class of computation.



- [1] H. Goforth and S. Lucey, *GPS-Denied UAV Localization using Pre-existing Satellite Imagery*, CMU Robotics Institute.  
<https://publications.ri.cmu.edu/gps-denied-uav-localization-using-pre-existing-satelliteimagery>  
(code: <https://github.com/hmgoforth/gps-denied-uav-localization>)
- [2] *ngps\_flight*: real-time CLIP-style UAV-satellite matching, ~3 m RMSE on Jetson Orin.  
[https://github.com/snktshrma/ngps\\_flight](https://github.com/snktshrma/ngps_flight)
- [3] *CLIP-UAV-localization*: CLIP-based network matching UAV imagery, also recovers heading.  
<https://github.com/codebra721/CLIP-UAV-localization>
- [4] *GNSS-Denied-UAV-Geolocalization*: terrain-weighted optimization using satellite maps and DEM data, sub-7 m.  
<https://github.com/YFS90/GNSS-Denied-UAV-Geolocalization>
- [5] *VisionUAV-Navigation*: classical SIFT/ORB hybrid feature detection, no deep learning.  
<https://github.com/sidharthmohannair/VisionUAV-Navigation>
- [6] GNSS-denied UAV geolocalization via terrain-weighted constraint optimization, sub-7 m error, *ScienceDirect*.  
<https://www.sciencedirect.com/science/article/pii/S1569843224006332>
- [7] CLIP-based UAV-satellite image matching, also derives heading direction, *ScienceDirect*.  
<https://www.sciencedirect.com/science/article/pii/S2590123025041787>







- [8] A Review of Key Technologies in Gravity Matching Navigation, *Sensors* (2026).  
<https://www.mdpi.com/1424-8220/26/13/4208>
- [9] Gravity-Aided Navigation Using Viterbi Map Matching Algorithm. <https://arxiv.org/abs/2204.10492>
- [10] Absolute Positioning Using the Earth's Magnetic Anomaly Field. <https://doi.org/10.1002/navi.138>
- [11] MagNav-Based Navigation Simulation Using Magnetic Anomaly Maps for GPS-Denied Environments.  
<https://doi.org/10.5302/J.ICRDS.2026.25.0344>
- [12] *PCVLabDrone2021*: real-time down-facing camera pipeline with quad-tree map retrieval, tested live.  
<https://github.com/OSUPCVLab/PCVLabDrone2021>
- [13] *geolocalization*: dual/Siamese network recognizing same-scene UAV and satellite image pairs.  
<https://github.com/abhinavtripathi95/geolocalization>
- [14] X. Zhu et al., *TransGeo: Transformer Is All You Need for Cross-view Image Geo-localization*, CVPR 2022.  
<https://arxiv.org/abs/2204.00097>
- [15] *FSRA*: A Transformer-Based Feature Segmentation and Region Alignment Method for UAV-View Geo-Localization.  
<https://arxiv.org/pdf/2201.09206>

